

# An Autofocus Algorithm for Estimating Residual Trajectory Deviations in Synthetic Aperture Radar

Lei Ran, Zheng Liu, Lei Zhang, Tao Li, and Rong Xie

**Abstract**—Due to the accuracy limitation of the navigation system, deviations between the real trajectory and the measured one appear inevitably in airborne synthetic aperture radar (SAR), which degrades the image quality dramatically. To improve the focusing performance, these trajectory deviations should be well estimated and compensated. In this paper, a data-based autofocus approach is proposed to correct the residual 3-D trajectory deviations. This new approach mainly contains two processing stages. The first stage is the local phase error estimation procedure involving small images autofocusing. A gradient function considering smoothness regularization is developed to efficiently achieve the sharpness-maximizing local phase error functions. In the second stage, the local phase error functions are combined to retrieve the residual 3-D trajectory deviations by a proposed weighted total least square method. This approach has been applied on highly squinted and large-swath airborne SAR raw data, respectively. Both real data experiments generate well-focused SAR images by the estimated trajectory parameters, and thus, validate the effectiveness of the proposed autofocus approach.

**Index Terms**—Autofocus, local phase error function, synthetic aperture radar (SAR), trajectory deviations, weighted total least square (WTLS) method.

## I. INTRODUCTION

SYNTHETIC aperture radar (SAR) has been widely applied as a remote sensing system in both military and civilian use. By transmitting wideband signals in a constant time interval, SAR can achieve the backscattering characteristics from the echoed data. To generate high-resolution microwave images of the observed area, SAR system usually is mounted on-board a platform, which flies at constant velocity along a nominal trajectory. Due to the atmospheric turbulence and the platform vibration, perfectly linear synthetic aperture array can hardly be formed for airborne SAR [1], [2]. In practical application, the trajectory parameters are usually measured by global positioning system and inertial measurement unit (IMU), and motion compensation (MOCO) is performed during the imaging processing. To achieve focused image, the measurement errors should be kept within fractions of a wavelength [1], [3], [5]. For high radar bands (e.g., X-band and Ku-band), this demand challenges the capacity of

some economic navigation system. In this case, the measured trajectory data only can be used for coarse MOCO, such as removal of the nonsystematic range cell migration and partial phase errors [4], [5]. After coarse MOCO, the image is still defocused, and residual motion errors should be estimated using the data-based autofocus technique.

Up to now, numerous autofocus methods have been developed for airborne SAR imagery. In conventional autofocus method, such as map-drift (MD) [4], [5] and phase gradient autofocus (PGA) [6]–[8], the phase errors are assumed invariant across the whole image. This limits their application in the presence of space variant phase errors. To alleviate the variant effect, autofocus strategy often involves sectioning the full image or data into small blocks, allowing the MD and PGA algorithms to be patchwise applied [4], [9]. The block size and the continuity of the phase error function should be elaborately considered to guarantee the compensation precision. Another category of autofocus is the metric-based method by maximizing or minimizing a particular object function. Popular metrics include image entropy [10], [11], sharpness [12]–[15], and contrast [16]. Although the metric-based autofocus method performs well on a variety of imagery and motion errors, it is usually time-consuming if the solution space is left completely unparametrized. Many efficient approaches have been proposed by reducing the solution space to several parameters, such as limited polynomial coefficients [10], [14] and discrete cosine transform (DCT) coefficients [11], [17]. However, as the aperture track is not updated in all aforementioned autofocus methods, the compensation is incomplete.

Recently, autofocus based on trajectory adjustment has been gaining more and more attention [18]–[25]. These autofocus techniques are mainly based on the aforementioned MD, PGA, and metric-based scheme. According to the variation of incidence angle within the radar beam, the trajectory errors can be estimated in stripmap PGA [18], and thus, the range-dependent phase errors are compensated. Li *et al.* [19] adopt the same motion error model in [18] and improve the trajectory estimate precision by filtering the PGA outputs and using the weighted total least square (WTLS) method. Considering the presence of squint angle, the motion error model in [18], [19] is modified in [20] to make it suitable for squint imaging mode. We notice that these autofocus methods [18]–[20] can provide robust estimation for cross-track motion error, but the deviation in along-track dimension remains uncompensated. For platform with small size and light weight, such as the unmanned aerial vehicle (UAV), the forward acceleration becomes noticeable [27]. Besides, in highly squinted SAR, the defocusing caused by the forward

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acceleration will be more pronounced than the broadside mode. Generally, the forward acceleration can be compensated via a proper resampling of the echoed signal in azimuth dimension [5], [19]. However, the resampling operation is time-consuming as it should be performed for all received data. The multilateration autofocus method is proposed in [21] to estimate the trajectory parameters using local MD algorithm. The trajectory deviations in [21] are updated by least square kernel; however, the accuracy of least square solution may be degraded by the estimation error of local MD algorithm. Factorized geometrical autofocus (FGA) [22], [23] can correct space variant RCM by assessing different trajectory parameters iteratively. The 6-D geometry search of FGA results in heavy computational burden, which limits its practical application. Recently, an efficient solution [24] has been established for FGA; however, the solution space still remains up to 6-D. The work in [25] demonstrates a less-memory and high-efficiency autofocus method to correct the trajectory by solving a system of nonlinear equations; however, the space variance of motion errors is neglected, which makes the updated trajectory only suitable for small scene imaging.

In this paper, we propose an improved MOCO approach aiming to estimate the trajectory deviations in both cross-track and along-track dimensions for airborne SAR imagery. This method mainly contains a two-stage processing. In the first stage, local images at different illumination directions within the radar beam are synthesized using back-projection (BP) algorithm. Maximum-sharpness phase-correction based on DCT-coefficients is utilized to derive the local phase error function (LPEF) for the blurred local image. To avoid local maxima [9], [26], a gradient expression by adding smoothness regularization is developed to achieve the sharpness-maximizing solution efficiently. The second stage performs a local-to-global computation according to the geometry relationship between the trajectory deviations and LPEFs. A WTLS method is proposed to improve the estimate accuracy in this stage. With the derived trajectory, the whole image can be finally focused. This new autofocus algorithm is suitable for airborne spotlight SAR imagery; however, it also can be applied for the stripmap mode using sub-aperture data.

The rest of this paper is organized as follows. In Section II, the motion error model is presented, and the time-domain SAR imaging related to this paper is described briefly. Section III deduces the new MOCO approach in detail. Section IV discusses some practical issues for its application. The flowchart of the new method is summarized in Section V. In Section VI, highly squinted and large-swath SAR data are used to demonstrate the performance of the proposed MOCO, respectively. And this paper is concluded in Section VII.

## II. MOTION ERROR MODEL AND TIME-DOMAIN SAR IMAGING

### A. Motion Error Model

The SAR geometry is shown in Fig. 1, where the straight line (parallel to  $x$ -axis) is the nominal trajectory, and the curved one denotes the real trajectory. In the ideal case, the platform moves along the nominal path with a constant

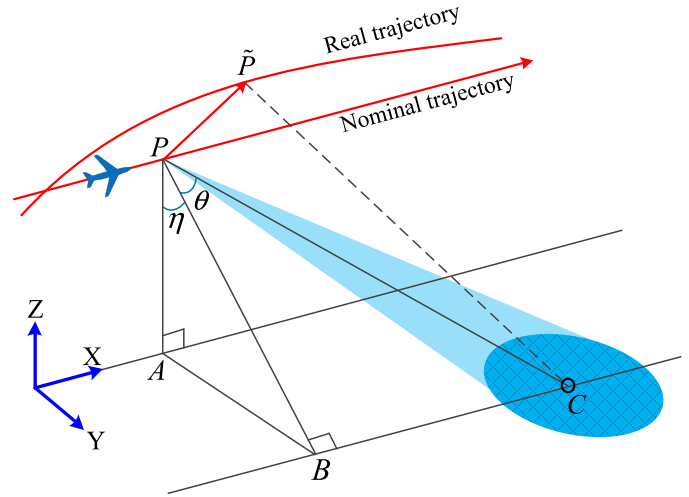


Fig. 1. Airborne SAR geometry with motion errors.

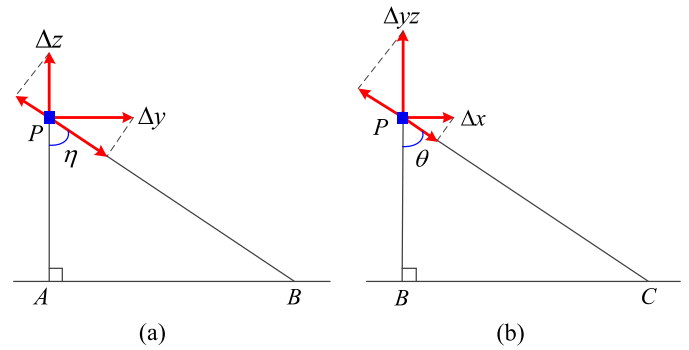


Fig. 2. Decomposition of the trajectory deviations. (a) PAB plane. (b) PBC plane.

velocity  $v$  and reference height  $H$ . However, the atmospheric turbulence usually makes the antenna phase center (APC) deviate from the ideal position in airborne SAR, yielding range errors for the targets in the image plane. Suppose that the ideal sensor position at slow time  $t_m$  is  $(vt_m, 0, H)$ , and the instantaneous motion error vector is  $[\Delta x(t_m), \Delta y(t_m), \Delta z(t_m)]$ , where  $\Delta x(t_m)$  is the along-track motion error, and  $\Delta y(t_m)$  and  $\Delta z(t_m)$  denote the cross-track displacements. Taking the motion error into account, the slant range from the sensor position to a target  $p$  at  $(x, y, z)$  can be written as

$$\begin{aligned} \tilde{R}_p(t_m) &= \sqrt{(vt_m + \Delta x(t_m) - x)^2 + (\Delta y(t_m) - y)^2 + (H + \Delta z(t_m) - z)^2} \\ &= R_p(t_m) + \Delta R_p(t_m) \end{aligned} \quad (1)$$

where  $R_p(t_m) = ((vt_m - x)^2 + y^2 + (H - z)^2)^{1/2}$  is the ideal slant range, and  $\Delta R_p(t_m)$  stands for the range error stemming from APC deviations. The motion error vector  $[\Delta x(t_m), \Delta y(t_m), \Delta z(t_m)]$  causes various defocusing effect for different SAR applications. For example, the blurs resulted by along-track deviations will be more pronounced in highly squinted mode than non-squint imaging; and the space variance of motion errors cannot be neglected for large swath SAR reconstruction since  $\Delta R_p(t_m)$  varies with the target position.

To make the range error coincide with the line-of-sight (LOS) direction, the motion error vector is decomposed according to the SAR geometry, as shown in Fig. 2.

The cross-track motion error is first decomposed in the normal plane  $PAB$  according to the incident angle  $\eta$ , which leads to  $\Delta yz(t_m) = \Delta y(t_m) \cdot \sin \eta + \Delta z(t_m) \cdot \cos \eta$ , ignoring the component vertical to  $\mathbf{PB}$ . In the slant range plane  $PBC$ , by projecting  $\Delta x(t_m)$  and  $\Delta yz(t_m)$  onto  $\mathbf{PC}$  direction, the LOS displacement can be computed by  $\Delta R(t_m) = \Delta x(t_m) \cdot \sin \theta(t_m) + \Delta yz(t_m) \cdot \cos \theta(t_m)$ , where  $\theta(t_m)$  is the instantaneous squint angle. The error perpendicular to  $\mathbf{PC}$  is discarded again as it makes no contribution for focusing. As a result, the range error in LOS direction for target  $\mathbf{p}$  is given by

$$\begin{aligned} \Delta R_p(t_m) = & \Delta x(t_m) \cdot \sin \theta_p(t_m) + \Delta y(t_m) \cdot \sin \eta_p(t_m) \\ & \cdot \cos \theta_p(t_m) + \Delta z(t_m) \cdot \cos \eta_p(t_m) \cdot \cos \theta_p(t_m). \end{aligned} \quad (2)$$

Since all trajectory deviations are accounted in (2), the precision of the motion error model is guaranteed. Besides, due to the linearized relationship between deviation and range error, the compensation can be performed efficiently, especially without the exhaustive azimuth resampling for along-track error correction.

To retrieve the variation information of motion error with the instantaneous squint angle and incident angle, we resort to the time-domain SAR processing [28]–[30]. A brief review of time-domain SAR imaging is given in the following.

### B. Time-Domain SAR Processing

Global back-projection (GBP) is a time-domain algorithm, projecting the range-compressed echoes to a pre-defined image plane in Cartesian coordinate or polar coordinate [29], [30]. The interpolated values from each pulse are added coherently, yielding reflective structures. GBP is a versatile algorithm; however, it is normally restricted to moderately sized image due to the number of operation, which is proportional to  $N^3$  for  $N$  pulses and an  $N \times N$  image.

Fast factorized back-projection (FFBP) [30] is a time-efficient alternative, combining the compressed echoes in a recursive manner. The basic principle is that FFBP partitions the GBP integral into subaperture images merging. The first stage of FFBP is the GBP formation, but using subaperture data and generating low-resolution polar sub-image. In the following stages of FFBP, low-resolution subimages are merged, resulting finer angular resolution and increased subaperture length. This merging procedure involves interpolation in both range and angle dimensions. Proper interpolator should be used to make up the fact that interpolation errors are accumulated in each factorized step. By a repeated recursion manner, full-resolution image finally can be obtained, blending the whole aperture data together. To avoid azimuth ambiguity, the angular interval of the polar image is restricted by

$$\Delta(\cos \xi^{(g)}) \leq \frac{\lambda_{\min}}{2L^{(g)}} \quad (3)$$

where  $\xi^{(g)}$  is the angle of polar grids in the  $g$ th processing stage,  $L^{(g)}$  denotes the subaperture length, and  $\lambda_{\min}$  represents the minimum wavelength of the transmitted signal.

The number of operations for a based-two FFBP implementation is proportional to  $2N^2 \log_2 N$  for  $N$  pulses and

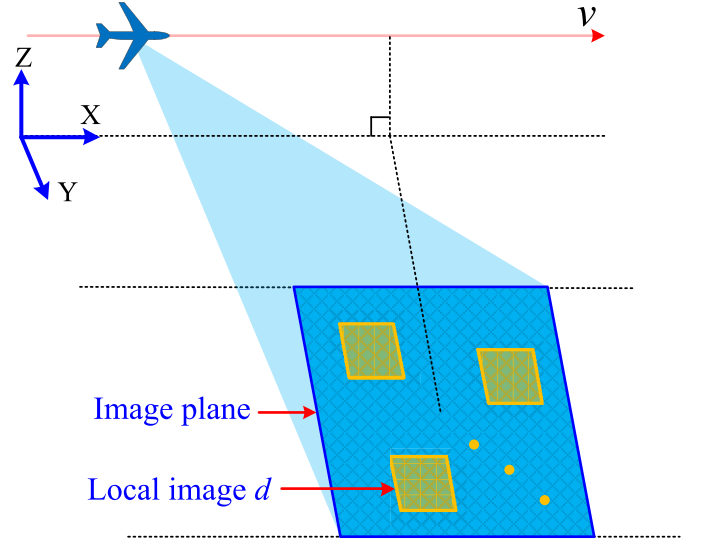


Fig. 3. Multiple local images selection.

$N \times N$  pixels, under the condition that  $N$  equals a power of two. FFBP dramatically reduces the computation burden of GBP, but retains the imaging precision.

GBP and FFBP can generate focused image with arbitrary flight track. However, their performance degrades seriously due to the unknown deviations. In the following, accurate trajectory parameters are estimated by the proposed MOCO method, using GBP for local image computation and FFBP for whole scene reconstruction.

## III. PROPOSED METHOD

Based on the motion error model in (2), the variation of range error with the projection angle should be derived to recover the unknown APC deviations. Since GBP gives the detailed geometry information between each acquisition position and the pixel of interest, local images at independent directions within the antenna beam are first synthesized using GBP, as shown in Fig. 3.

### A. Local Phase Error Estimation

Assume that  $D$  small scenes with equal size are selected, and the  $d$ th local image with motion error is computed by

$$\begin{aligned} \tilde{\mathbf{x}}_d &= \sum_{n=1}^N \sigma_d(n) \cdot \exp \left[ -j \frac{4\pi}{\lambda} \tilde{\mathbf{R}}_d(n) \right] \cdot \exp \left[ j \frac{4\pi}{\lambda} \mathbf{R}_d(n) \right] \\ &= \sum_{n=1}^N \sigma_d(n) \cdot \exp \left[ -j \frac{4\pi}{\lambda} \Delta \mathbf{R}_d(n) \right] \end{aligned} \quad (4)$$

where the subscript  $d$  is the index of local images,  $\sigma_d(n)$  is the back-projected vector of the  $n$ th ( $n = 1, 2, \dots, N$ ) pulse covering all pixels in local image,  $\tilde{\mathbf{R}}_d(n)$  denotes the slant range vector with motion error, while  $\mathbf{R}_d(n)$  represents the ideal one,  $\Delta \mathbf{R}_d(n)$  contains the range errors for pixels of interest, and  $\lambda$  is the wavelength corresponding to the carrier frequency. Due to the small size of individual local image, the pixel-dependence of  $\Delta \mathbf{R}_d(n)$  in (4) can be neglected,

which means the  $d$ th LPEF is space invariant. Let  $\varphi_d(n) = -(4\pi/\lambda)\Delta\mathbf{R}_d(n)$  denote the phase error related to the  $n$ th pulse, and then, (4) can be simplified as

$$\tilde{\chi}_d = \sum_{n=1}^N \sigma_d(n) \cdot \exp[j\varphi_d(n)] = \sum_{n=1}^N \tilde{\sigma}_d(n) \quad (5)$$

where  $\tilde{\sigma}_d(n)$  is the phase-corrupted back-projection vector. In the following, the blurred local image is ready to be refocused by maximizing a quality metric, i.e., sharpness [15].

To generate high-resolution image, the synthetic aperture usually collects thousands of pulses, which means that the LPEF, i.e.,  $\varphi_d = [\varphi_d(1), \varphi_d(2), \dots, \varphi_d(N)]_{1 \times N}$ , consists of thousands of unknown parameters. It is difficult to solve this problem because of the large solution space and high computation burden. Fortunately, owing to the inertia of airborne SAR, the LPEF stemming from motion error is smooth and continuous. As a consequence, LPEF can be expressed by a polynomial function with limited orders [10], [14] or an inverse DCT of several DCT-coefficients [11]. Both expressions can reduce the solution dimension dramatically, thus minimizing the computational load. The limited polynomials mainly extract the low frequency part of phase errors, while DCT-coefficients can represent the low- and high-frequency components of LPEF simultaneously. Therefore, the LPEF is estimated based on DCT-coefficients in this paper.

The approximate expression of  $\varphi_d$  based on DCT-coefficients is given by

$$\tilde{\varphi}_d(n) = \sum_{k=0}^{N-1} a(k) \cdot \zeta_d(k) \cdot \cos\left[\frac{\pi k(2n+1)}{2N}\right] \quad (6)$$

where

$$a(k) = \begin{cases} 1/\sqrt{N} & (k=0) \\ \sqrt{2/N} & (k \neq 0) \end{cases}$$

$$\begin{cases} \zeta_d(k) \neq 0 & (k=0, 1, \dots, N_{\text{DCT}}-1) \\ \zeta_d(k) = 0 & (k=N_{\text{DCT}}, N_{\text{DCT}}+1, \dots, N-1) \end{cases}$$

$\zeta_d(k)$  is the  $k$ th DCT-coefficient,  $N_{\text{DCT}}$  denotes the number of the preserved DCT-coefficients, and  $N_{\text{DCT}} < N$ . Compared with the original  $\varphi_d$ , the number of undetermined parameters in  $\tilde{\varphi}_d$  reduces from  $N$  to  $N_{\text{DCT}}$ , which allows high efficiency. In practical, 30-50 orders will be accurate enough for typical airborne SAR platform movement.

Using (6), the phase-corrected local image becomes

$$\tilde{\chi}_d = \sum_{n=1}^N \tilde{\sigma}_d(n) \cdot \exp[-j\tilde{\varphi}_d(n)] = \sum_{n=1}^N \hat{\sigma}_d(n). \quad (7)$$

The sharpness metric involving motion error for the  $d$ th local image is computed by

$$f(\tilde{\varphi}_d) = \sum_{p=1}^P [\hat{\chi}_d(p) \cdot \hat{\chi}_d^*(p)]^2 \quad (8)$$

where  $P$  denotes the total number of pixels in the  $d$ th local image and  $(\cdot)^*$  the conjugate operator. The problem to estimate

the LPEF can be described by maximizing the sharpness metric

$$\hat{\varphi}_d = \arg \max_{\tilde{\varphi}_d} f(\tilde{\varphi}_d). \quad (9)$$

The more  $\hat{\varphi}_d$  approaches to the actual LPEF, the better focusing performance the MOCO can provide.

To retrieve the LPEF under the criterion of maximum sharpness, an efficient way is to express the gradient of the metric as a function of the undetermined DCT-coefficients, and then gradient descent algorithm can be used to find the optimal solution. Using the chain rule [9], the gradient of sharpness cost with respect to the  $k$ th DCT-coefficient is as follows:

$$\begin{aligned} \frac{\partial f(\tilde{\varphi}_d)}{\partial \zeta_d(k)} &= \frac{\partial f(\tilde{\varphi}_d)}{\partial \hat{\chi}_d} \cdot \frac{\partial \hat{\chi}_d}{\partial \tilde{\varphi}_d} \cdot \frac{\partial \tilde{\varphi}_d}{\partial \zeta_d(k)} \\ &= 2 \sum_{p=1}^P [\hat{\chi}_d(p) \cdot \hat{\chi}_d^*(p)] \\ &\quad \times \left[ \hat{\chi}_d^*(p) \cdot \frac{\partial \hat{\chi}_d(p)}{\partial \zeta_d(k)} + \hat{\chi}_d(p) \cdot \frac{\partial \hat{\chi}_d^*(p)}{\partial \zeta_d(k)} \right]. \end{aligned} \quad (10)$$

And the gradient of the  $d$ th local image in terms of DCT-coefficient is calculated by

$$\begin{aligned} \frac{\partial \hat{\chi}_d(p)}{\partial \zeta_d(k)} &= \frac{\partial \hat{\chi}_d(p)}{\partial \tilde{\varphi}_d} \cdot \frac{\partial \tilde{\varphi}_d}{\partial \zeta_d(k)} \\ &= -j \cdot \sum_{n=1}^N \left\{ \hat{\sigma}_d(n; p) \cdot a(k) \cdot \cos\left[\frac{\pi k(2n+1)}{2N}\right] \right\} \end{aligned} \quad (11)$$

where  $\hat{\sigma}_d(n; p)$  is the phase-corrected back-projection value for the  $p$ th pixel at the  $n$ th sensor position. Meanwhile, the conjugate form  $(\partial \hat{\chi}_d^*(p)/\partial \zeta_d(k))$  is given accordingly

$$\begin{aligned} \frac{\partial \hat{\chi}_d^*(p)}{\partial \zeta_d(k)} &= \frac{\partial \hat{\chi}_d^*(p)}{\partial \tilde{\varphi}_d} \cdot \frac{\partial \tilde{\varphi}_d}{\partial \zeta_d(k)} \\ &= j \cdot \sum_{n=1}^N \left\{ \hat{\sigma}_d^*(n; p) \cdot a(k) \cdot \cos\left[\frac{\pi k(2n+1)}{2N}\right] \right\}. \end{aligned} \quad (12)$$

Substituting (7), (11) and (12) into (10), the gradient formula can be obtained after some rearrangement

$$\begin{aligned} \frac{\partial f(\tilde{\varphi}_d)}{\partial \zeta_d(k)} &= 4 \sum_{p=1}^P \left\{ \left| \sum_{n=1}^N \hat{\sigma}_d(n; p) \right|^2 \times \text{Im} \left\{ \sum_{n=1}^N \hat{\sigma}_d^*(n; p) \right. \right. \\ &\quad \left. \left. \cdot \sum_{n=1}^N \left\{ \hat{\sigma}_d(n; p) \cdot a(k) \cdot \cos\left[\frac{\pi k(2n+1)}{2N}\right] \right\} \right\} \right\} \end{aligned} \quad (13)$$

where  $|\cdot|$  is the absolute operator, and  $\text{Im}(\cdot)$  denotes the imaginary part of the complex value.

With the gradient expression in (13), the first-order derivatives of image sharpness with respect to DCT-coefficients are

$$\nabla f(\tilde{\phi}_d) = \left[ \frac{\partial f(\tilde{\phi}_d)}{\partial \zeta_d(0)}, \frac{\partial f(\tilde{\phi}_d)}{\partial \zeta_d(1)}, \dots, \frac{\partial f(\tilde{\phi}_d)}{\partial \zeta_d(N_{\text{DCT}} - 1)} \right]_{1 \times N_{\text{DCT}}} \quad (14)$$

Many gradient descent algorithms can be applied to find the sharpness-maximizing solution. However, suboptimal solution may be produced due to the fact that the gradient descent search converges toward local sharpness-maximum [9], [26], especially when proper initial condition is not available for each local image. From our experience, the gradient descent algorithm is easily trapped in a local maximum in the real data experiment, which ruins the estimate precision and the convergency speed.

In order to dodge this problem, smoothness regularization can be used to penalize the searching direction [9]. Owing to the inertia of airborne SAR platform, the LPEF induced by motion error is smooth and continuous [4], [5], [9], [27], which yields the fact that the low-frequency components are more dominated than the high-frequency ones in DCT-coefficients distribution. This can be utilized to bias the gradient searching direction during the iterations. The gradient expression with smoothness regularization is developed as follows:

$$\nabla f_{\text{reg}}(\tilde{\phi}_d) = \nabla f(\tilde{\phi}_d) \odot \Omega_d \quad (15)$$

where  $\Omega_d \in \mathbb{R}^{1 \times N_{\text{DCT}}}$  denotes the weighting vector ranging from 0 to 1 for the  $d$ th local image, and  $\odot$  represents the Hadamard product. In (15), the contribution of different gradients to the searching direction is weighted by  $\Omega_d$ , attempting to find the global maximum solution. As mentioned before, the gradients corresponding to the low-frequency DCT-coefficients should be encouraged by large weights, while the ones with respect to high-frequency components penalized by small weights.

Here comes the immediate question: how to determine the weighting vector  $\Omega_d$ ? One possible way could be the usage of IMU data from the navigation system. Despite the fact that the IMU data are not accurate enough for focusing, it shows the trend of the platform motion. Accordingly, the imperfect LPEF calculated by IMU data inherits the variation characteristic of the trajectory. Therefore, DCT results of the imperfect LPEF can provide proper prior knowledge for the weighting vector. To some extent, the IMU-based weighting strategy may not be accurate enough to represent the frequency characteristic of the real LPEF; however, they are helpful to avoid local maximum and accelerate the convergency. The gradient descent algorithm can make up the inaccuracy of the weighting vector, thus finding the precisely sharpness-maximizing solution.

We apply the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm [9], [32] to estimate the LPEFs in this paper. The BFGS method is a very effective algorithm to solve the unconstrained, nonlinear optimization problem with multiple variables, and has the potential to converge at a superlinear rate [32]. The BFGS search considering smoothness regularization for the  $d$ th LPEF includes the following steps:

*Step 1:* Initialization. Set the iterative number  $u = 1$ , the initial solution  $\hat{\phi}_d^{[u]} = \mathbf{0}$ , the regularization vector  $\Omega_d$ , and a threshold  $\varsigma_T$  used to terminate the algorithm.

*Step 2:* Calculate the gradient expression  $\nabla f(\hat{\phi}_d^{[u]})$  using (10)–(14).

*Step 3:* Get the regularized gradient expression  $\nabla f_{\text{reg}}(\hat{\phi}_d^{[u]})$  by (15).

*Step 4:* Find the descent step-length  $\alpha_d^{[u]}$  according to

$$\begin{aligned} f[\hat{\phi}_d^{[u]} + \alpha_d^{[u]} \cdot (\mathbf{B}_d^{[u]})^{-1} \cdot \nabla f_{\text{reg}}(\hat{\phi}_d^{[u]})] \\ \geq f(\hat{\phi}_d^{[u]}) + \alpha_d^{[u]} \cdot [\nabla f_{\text{reg}}(\hat{\phi}_d^{[u]})]^T \cdot (\mathbf{B}_d^{[u]})^{-1} \cdot \nabla f_{\text{reg}}(\hat{\phi}_d^{[u]}) \end{aligned} \quad (16)$$

where  $\mathbf{B}$  denotes the estimated Hessian matrix [32], and  $(\cdot)^T$  and  $(\cdot)^{-1}$  represent the matrix/vector transpose and inverse, respectively.

*Step 5:* Estimate the LPEF, which leads to

$$\hat{\phi}_d^{[u+1]} = \hat{\phi}_d^{[u]} + \alpha_d^{[u]} \cdot (\mathbf{B}_d^{[u]})^{-1} \cdot \nabla f_{\text{reg}}(\hat{\phi}_d^{[u]}). \quad (17)$$

*Step 6:* Remove the linear phase part in  $\hat{\phi}_d^{[u+1]}$ .

*Step 7:* Repeat steps 2–6 until satisfying the termination condition  $f(\hat{\phi}_d^{[u+1]}) - f(\hat{\phi}_d^{[u]}) < \varsigma_T$ , generating the final sharpness-maximizing LPEF  $\hat{\phi}_d = \hat{\phi}_d^{[u+1]}$ .

Specifically, details about the derivation of the Hessian matrix  $\mathbf{B}$  are presented in Appendix. In addition, the linear phase perturbation has been taken into account in **Step 6**. For practical application, linear phase can be eliminated through Doppler centroid estimation [5], [31]. However, different linear phase parts are usually involved in different LPEFs from our experience. To deal with this problem, linear phase of individual LPEF is removed during the BFGS iterations using polynomial curve fitting method [27], as illustrated in Step 6. Notably, in GBP formation, the linear phase not only causes image shift, but also results in focusing degradation, which is significantly different from the frequency imaging algorithm [19], [27]. Removing the linear phase may influence the sharpness value, but such effect will be less and less noticeable in successive iterations.

We found that adding smoothness regularization to the gradient formula is a necessary step to accelerate the convergency speed and avoid local maximum, especially in the early iterations of BFGS search. Detailed examples for applying the regularized gradient expression will be presented in the real data experiments in Section VI.

### B. Residual Trajectory Deviations Calculation

After all selected local images are autofocused using the BFGS method,  $D$  LPEFs and  $D$  focused local images are now available for further trajectory deviations estimation. According to the projection relationship between the 3-D deviations and the SAR geometry, the derived LPEFs can be combined into a system of linear equations to estimate the residual trajectory parameters [21].

In accordance with (2), the discrete form of the motion error model for the  $d$ th local image at the  $n$ th sensor position is

$$\mathbf{M} = \begin{bmatrix} \sin \bar{\theta}_1(n) & \sin \bar{\eta}_1(n) \cdot \cos \bar{\theta}_1(n) & \cos \bar{\eta}_1(n) \cdot \cos \bar{\theta}_1(n) \\ \sin \bar{\theta}_2(n) & \sin \bar{\eta}_2(n) \cdot \cos \bar{\theta}_2(n) & \cos \bar{\eta}_2(n) \cdot \cos \bar{\theta}_2(n) \\ \vdots & \vdots & \vdots \\ \sin \bar{\theta}_D(n) & \sin \bar{\eta}_D(n) \cdot \cos \bar{\theta}_D(n) & \cos \bar{\eta}_D(n) \cdot \cos \bar{\theta}_D(n) \end{bmatrix}_{D \times 3} \quad (18)$$

given as follows:

$$\begin{aligned} \Delta \hat{R}_d(n) &= \Delta x(n) \cdot \sin \bar{\theta}_d(n) + \Delta y(n) \cdot \sin \bar{\eta}_d(n) \cdot \cos \bar{\theta}_d(n) \\ &\quad + \Delta z(n) \cdot \cos \bar{\eta}_d(n) \cdot \cos \bar{\theta}_d(n) \end{aligned} \quad (19)$$

where  $\Delta \hat{R}_d \in \mathbb{R}^{1 \times N}$  is the  $d$ th spatial-invariant range error function, which is calculated by  $\Delta \hat{R}_d = (\lambda/4\pi) \cdot \hat{\phi}_d$ , and  $\bar{\theta}_d \in \mathbb{R}^{1 \times N}$  and  $\bar{\eta}_d \in \mathbb{R}^{1 \times N}$  denote the equivalently instantaneous squint angle and incident angle, respectively.

Because that local image instead of single pixel is used to estimate the LPEF,  $\bar{\theta}_d$  and  $\bar{\eta}_d$  should be synthesized by the whole local image rather than a single pixel. As the GBP gives perfect projection geometry for individual pixel, the equivalent  $\bar{\theta}_d$  and  $\bar{\eta}_d$  can be calculated as follows:

$$\bar{\theta}_d(n) = \sum_{i=1}^{I_d} \frac{[\hat{\chi}_d(i) \cdot \hat{\chi}_d(i)^*]^2 \cdot \theta_d(i; n)}{f(\hat{\chi}_d)} \quad (20)$$

$$\bar{\eta}_d(n) = \sum_{i=1}^{I_d} \frac{[\hat{\chi}_d(i) \cdot \hat{\chi}_d(i)^*]^2 \cdot \eta_d(i; n)}{f(\hat{\chi}_d)}. \quad (21)$$

In (19) and (20),  $\hat{\chi}_d = \sum_{n=1}^N \bar{\sigma}_d(n) \cdot \exp[-j\hat{\phi}_d(n)]$  is the  $d$ th LPEF-corrected local image,  $f(\hat{\chi}_d)$  is the sharpness of  $\hat{\chi}_d$ ,  $\theta_d(i; n)$  and  $\eta_d(i; n)$  denote the ideal squint angle and incident angle between the  $i$ th pixel and the  $n$ th acquisition position, respectively, and  $I_d$  represents the total number of pixels in the  $d$ th local image. The ratio of  $[\hat{\chi}_d(i) \cdot \hat{\chi}_d(i)^*]^2$  to  $f(\hat{\chi}_d)$  acts as weight for individual pixel, which yields the fact that the pixel contributing most to the sharpness metric plays the biggest role in the computation of synthetic squint angle and incident angle. Besides, the flat topography assumption is adopted to compute  $\theta_d(i; n)$  and  $\eta_d(i; n)$ .

Using (19) and (20), we get the projection matrix as shown at the top of the page.

For the  $n$ th acquisition position, the spatial-variant range errors from  $D$  local images are combined into the following vector:

$$\mathbf{P} = [\Delta \hat{R}_1(n) \quad \Delta \hat{R}_2(n) \quad \cdots \quad \Delta \hat{R}_D(n)]_{D \times 1}^T \quad (22)$$

Furthermore, the primed trajectory deviations are vectorized as

$$\mathbf{\Lambda} = [\Delta x(n) \quad \Delta y(n) \quad \Delta z(n)]_{3 \times 1}^T. \quad (23)$$

Then the motion error model can be rewritten into the following form:

$$\mathbf{P} = \mathbf{M}\mathbf{\Lambda} + \mathbf{e} \quad (24)$$

where  $\mathbf{e} = [\mathbf{e}_1(n) \quad \mathbf{e}_2(n) \quad \cdots \quad \mathbf{e}_D(n)]_{D \times 1}^T$ , and  $\mathbf{e}_d \in \mathbb{R}^{1 \times N}$  represents the estimation error in the  $d$ th LPEF, which mainly caused by the noise and clutter.

According to Gauss-Markov Theorem [33], if the estimation error from each LPEF is statistically independent and has same

variance, the least square (LS) method can be applied directly to get the optimal solution in the sense of minimum variance

$$\mathbf{\Lambda}_{LS} = \arg \min(\mathbf{e}^T \mathbf{e}) \quad (25)$$

which yields

$$\hat{\mathbf{\Lambda}}_{LS} = (\mathbf{M}^T \mathbf{M})^{-1} \mathbf{M}^T \mathbf{P}. \quad (26)$$

However, the assumption of same variance can hardly hold true in practice, especially for local images with different signal to noise ratios (SNRs). In this case, the LS method cannot provide minimum variance solution anymore, and  $\mathbf{\Lambda}$  should be estimated by the weighted LS (WLS) method

$$\mathbf{\Lambda}_{WLS} = \arg \min(\mathbf{e}^T \mathbf{W}^{-1} \mathbf{e}) \quad (27)$$

which yields

$$\hat{\mathbf{\Lambda}}_{WLS} = (\mathbf{M}^T \mathbf{W}^{-1} \mathbf{M})^{-1} \mathbf{M}^T \mathbf{W}^{-1} \mathbf{P} \quad (28)$$

where  $\mathbf{W} = \text{diag}(w_1^2, w_2^2, \dots, w_D^2)$ , and  $w_d^2 = \text{var}(\Delta \hat{R}_d)$ . The symbol  $\text{diag}(\cdot)$  denotes diagonal matrix and  $\text{var}(\cdot)$  represents the variance.

For practical application, errors may exist in LPEFs and projection matrices simultaneously, which means that both  $\mathbf{P}$  and  $\mathbf{M}$  contain errors. For instance, some parameters, such as the size of local images, the orders of DCT-coefficients, the interpolator kernel of GBP, and the numerical parameters of BFGS, have an influence on the accuracy of LPEFs. If inappropriate processing parameters are involved in sharpness-based optimization, the estimation errors are definitely brought into the LPEFs. On the other hand, the flat topography assumption is adopted for the calculation of projection matrix. This indicates that the instantaneous squint angle and incident angle will deviate from the real ones for various terrains. Considering both errors in  $\mathbf{P}$  and  $\mathbf{M}$ , (24) should be modified as follows:

$$\mathbf{P} = (\mathbf{M} - \mathbf{E}) \mathbf{\Lambda} + \mathbf{e} \quad (29)$$

where  $\mathbf{E} \in \mathbb{R}^{D \times 3}$  denotes the error matrix corresponding to  $\mathbf{M}$ . Both errors of  $\mathbf{E}$  and  $\mathbf{e}$  should be taken into account for precise trajectory correction.

The LS and WLS methods consider the errors in the observation, i.e.,  $\mathbf{P}$ . However, they fail to eliminate the effect of errors in  $\mathbf{M}$ , which makes the LS and WLS solutions inconsistent and biased in practice. In such condition, the total least square (TLS) method [34], [35] can be used to improve the estimate accuracy. TLS aims to solve the following problem:

$$\mathbf{\Lambda}_{TLS} = \arg \min (\|\mathbf{E}\|_F^2 + \mathbf{e}^T \mathbf{e}) \quad (30)$$

where  $\|\cdot\|_F$  denotes the Frobenius norm. The TLS solution to (30) is well documented in [34], [35]. First, the augmented

matrix  $[\mathbf{M} \ \mathbf{P}] \in \mathbb{R}^{D \times 4}$  is examined using singular value decomposition

$$[\mathbf{M} \ \mathbf{P}] = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \quad (31)$$

where  $\mathbf{\Sigma} \in \mathbb{R}^{D \times 4}$  contains the singular values, and  $\mathbf{U} \in \mathbb{R}^{D \times D}$  and  $\mathbf{V} \in \mathbb{R}^{4 \times 4}$  are the left and right singular matrices, respectively. Then the TLS estimator is given by

$$\hat{\mathbf{A}}_{\text{TLS}} = -\frac{1}{V(4, 4)} \cdot \begin{bmatrix} V(1, 4) \\ V(2, 4) \\ V(3, 4) \end{bmatrix} \quad (32)$$

where  $[V(1, 4), V(2, 4), V(3, 4), V(4, 4)]^T$  denotes the right singular vector corresponding to the smallest singular value.

As mentioned before, the variance of LPEFs is still not same, violating the homoscedastic error structure required by TLS. To retrieve the 3-D trajectory deviations from multiple LPEFs of varying accuracy, a WTLS method is proposed in this paper by optimizing the following problem

$$\mathbf{A}_{\text{WTLS}} = \arg \min (\|\mathbf{E}\|_F^2 + \mathbf{e}^T \mathbf{W}^{-1} \mathbf{e}) \quad (33)$$

where  $\mathbf{W} = \text{diag}(w_1^2, w_2^2, \dots, w_D^2)$ , and  $w_d^2 = \text{var}(\Delta \hat{\mathbf{R}}_d)$ . When establishing the WTLS model in (33), two significant factors are accounted. First, weighting matrix  $\mathbf{W}$  is used to control the contribution of individual LPEF to the computation of the optimal solution, which leads to the fact that the estimated trajectory deviations are less influenced by the LPEF with large error. As an excellent result, WTLS estimation will be more accurate and robust than the TLS method. Second, another factor we want to stress is that the weighting strategy is applied for  $\mathbf{e}$ , but not for  $\mathbf{E}$ . The errors in  $\mathbf{E}$  come from two aspects: the flat topography assumption and the calculation of equivalent squint angle and incident angle using (19) and (20). Obviously, the results of (19) and (20) are located within the local image. However, due to the small size of each local image, the related LPEF is assumed to be space-invariant, which means that any squint angle and incident angle within the local image can be treated as equivalent values. Therefore, heteroscedastic errors [36]–[38] will not be brought into the columns of  $\mathbf{E}$ . In consequence, the errors in  $\mathbf{E}$  are mainly caused by the flat topography assumption. Owing to the unknown terrain structure, it is hard to determine the variance characteristics of the errors in  $\mathbf{E}$ . Therefore, error matrix  $\mathbf{E}$  is considered as homoscedastic and remains unweighted in the proposed WTLS model. More sophisticated weighting schemes applied for  $\mathbf{E}$  and  $\mathbf{e}$  simultaneously may be investigated in our future work, if detailed terrain information is available.

Unlike the LS, WLS, and TLS methods, the WTLS estimator has no analytic closed-form solution in general case, and therefore numerical iterative optimization method is employed to find the desired trajectory deviations. In this paper, the residual trajectory deviations are solved using the iterative algorithm in [37], which is suitable for heteroscedastic observations, i.e., LPEFs, in our model.

The WTLS method reduces the sensitivity of the solution to numerical and estimation errors. Note that increasing the number of LPEFs by selecting more local images has beneficial effect on the accuracy of the WTLS solution; however,

the whole MOCO procedure would not be very efficient from a computational viewpoint. Therefore, it is better to make a tradeoff between accuracy and efficiency.

#### IV. PRACTICAL CONSIDERATIONS

Two processing stages, i.e., sharpness-based LPEFs estimation and WTLS-based trajectory deviations calculation, of the proposed MOCO method have been deduced theoretically in the previous section, but more details need to be discussed before we apply the new MOCO approach to the raw data experiments. Some practical considerations are now in the following.

##### A. Time-Domain Imaging Using Nonlinear Trajectory

In the proposed MOCO framework, GBP and FFBP algorithms are used to compute the local image and full-swath image, respectively, using nonlinear trajectory parameters. For local image synthesis via GBP, the distance between the acquisition position and the pixel on the image plane are calculated directly using nonlinear track parameters. Then the range-compressed echoes are upsampled eight times through zero padding in the range frequency domain, and nearest neighbor interpolation is performed on the upsampled data to form blurred local images. Note that, the back-projected values of each pulse should be stored in memory, which are used for the LPEF estimation. Due to the small size of local image, this will not result in heavy memory burden, but decreases the computational burden vastly. Suppose that residual 3-D trajectory deviations have been derived using the WTLS method, FFBP is then employed to generate the full-swath image. The first processing step of FFBP is similar to GBP, but using sub-aperture data and the estimated trajectory. In the following subimages recursion step, 2-D sinc interpolation is performed in both range and angle dimensions.

##### B. Local Image Selection

Because individual local image is autofocused by the sharpness-maximizing optimization, and the strong scattering point plays the biggest role to the computation of sharpness metric, it is advisable to choose local images containing strong points. To guarantee the space invariance of the LPEF, the local image size should be kept small; otherwise, the estimate precision and focusing performance will be degraded. Another defect for selecting large local image is the heavy computation burden and memory requirement [15], [25]. Conversely, if the local image is too small to cover the blurred power of the strong points, the sharpness-maximizing solution becomes unreliable as well. Therefore, the size of the local image should be designed elaborately. The last factor we want to emphasize is the relative location of the selected local images. On one hand, if the local images have comparable range, the incident angle for each local image remains nearly unchanged, which makes the last two columns of the projection matrix  $\mathbf{M}$  nearly correlated. In this case, it would be difficult to distinguish the motion information in the  $y$  and  $z$  directions. On the other hand, if the selections are centralized, both the instantaneous

squint angles and incident angles for the local images have little difference, which leads to the seriously ill-posed equation systems. This is not beneficial for generating reliable solution. As a result, local images should distribute over a wide extent, providing sufficiently spatial-variant information of the imaging geometry. For practice, local images containing strong scattering points, covering the most dispersed blurs and distributing over a wide extent are suggested for LPEFs estimation. Such selection helps to improve the estimate accuracy, accelerate the convergency speed, and ease the memory requirement. However, if the observed scenario is short of such selections, the performance of the proposed MOCO may degrade inevitably.

### C. Residual Range Cell Migration Correction

The proposed MOCO aims to estimate and correct phase error stemming from trajectory deviations, providing that there is no residual range cell migration (RCM) error in SAR echoes. Generally, RCM error can be corrected by the measured data from navigation system. However, for cases of high-resolution and measurement sensor with low accuracy, RCM error may even exit after coarse compensation by the measured trajectory. To eliminate the residual RCM error, strategies, such as downsampling in range frequency domain [21] and a coarse range spacing of the image grids are recommended [25]. Residual RCM is usually considered as spatial-invariant in airborne SAR, which indicates that we can select only one local image and use the sharpness-based BFGS search to find the residual phase error and the corresponding RCM error. To improve the autofocusing precision, this procedure may be performed in several iterations by gradually reducing the downsampling factor and decreasing the range spacing of the image grids. However, since the measured trajectory parameters provide reliable correction for the residual RCM in our succeeding raw data experiments, this extra autofocusing stage for residual RCM correction is not implemented in this study. For future situation that the accuracy of the measurement sensor challenges the proposed MOCO, residual RCM correction is necessary. We leave the residual RCM correction as an alternative choice for various airborne SAR data sets.

### D. Time Efficiency

In terms of the computational load of the two processing stages in the proposed MOCO, the sharpness-based LPEF estimation is more time-consuming than the WTLS calculation at the moment. Because the LPEF estimation is compatible with the GBP formation, it has higher computational burden than the conventional autofocus methods, such as MD and PGA, which can be implemented by the Fast Fourier Transform (FFT). However, selecting small local images and using the regularized gradient formula will help to find the optimal LPEF solution efficiently. In addition, the LPEF estimation for different local images and WTLS calculation covering all acquisition positions can be implemented in a parallelized way, which may further accelerate the computational speed, and make it suitable for real-time application [39], [40].

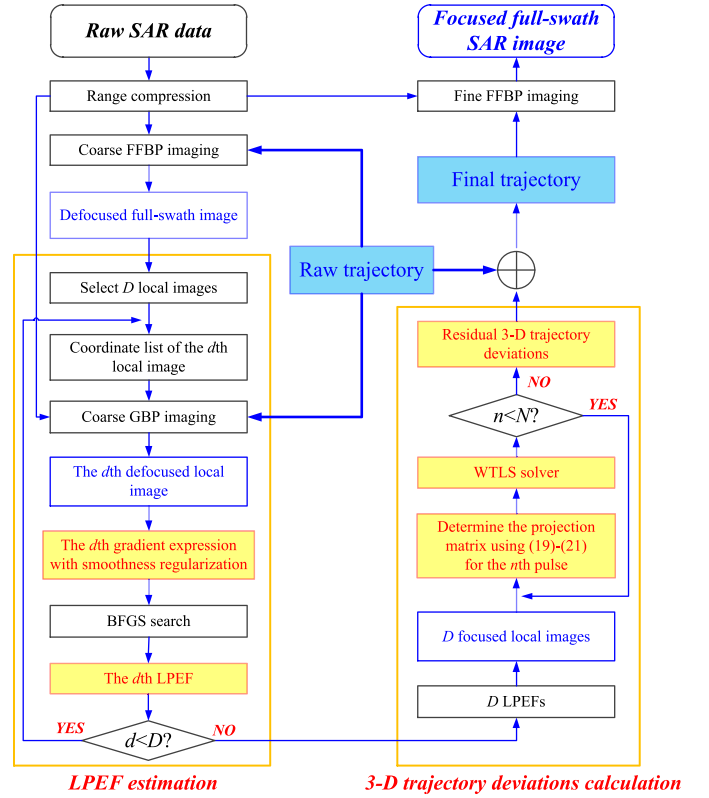


Fig. 4. Flowchart of the proposed MOCO method.

## V. FRAMEWORK OF THE PROPOSED METHOD

With the theoretical analysis and practical considerations discussed in the previous chapter, the flowchart of the proposed MOCO method is now summarized in Fig. 4. To implement the new method, the range-compressed echoes are first interpolated to generate a full-swath image using the measured trajectory. Based on this defocused full-swath image, multiple local images at independent illumination directions are re-synthesized using GBP processor. Then the blurred local images are autofocused under the criterion of maximum sharpness, yielding multiple LPEFs. According to the effect of the sensor position bias on the LPEFs, a linear system of equations is then set up and solved by a proposed WTLS kernel to retrieve the unrecorded trajectory deviations. Finally, accurate trajectory update can be achieved by adding the raw trajectory and the estimated one, and subsequently, focused full-swath image is available using the final trajectory. In practice, due to the complication of motion errors for different platforms, the proposed MOCO may be iterated more than once to improve the focusing precision.

One may notice that the FFBP imaging is performed twice in our MOCO flowchart. The first time generates the coarse image for local images selection, and the second one is used to obtain the refocused image. This would not be very efficient from a computational standpoint. We can set a relative larger grid spacing of the image plane to reduce the resolution in the first FFBP imaging, which has little impact on the local image selection, but improves the time-efficiency. Actually, the most important contribution of the new MOCO approach

TABLE I  
MAIN PARAMETERS OF BOTH REAL DATA SETS

| Parameter                         | Data set 1                | Data set 2                |
|-----------------------------------|---------------------------|---------------------------|
| Operational band                  | Ku                        | Ku                        |
| Squint angle                      | 50°                       | 0°                        |
| Bandwidth                         | 200MHz                    | 200MHz                    |
| Range (scene center)              | 7.3km                     | 9km                       |
| Mean altitude                     | 1.9km                     | 1.8km                     |
| Mean velocity                     | 50m/s                     | 55m/s                     |
| Aperture length                   | 393.2m                    | 450.7m                    |
| Image grid (range $\times$ angle) | 0.6m $\times$ 0.0038°     | 0.6m $\times$ 0.0032°     |
| Image size (range $\times$ angle) | 2800 $\times$ 1500 pixels | 6000 $\times$ 1600 pixels |

is to estimate the residual trajectory deviations. Once the corrected trajectory is available, the twice FFBP processing can be replaced by other high-efficient imaging algorithms, as long as they can deal with the nonlinear trajectory.

## VI. EXPERIMENTS

In the following, two real data sets are used to investigate the performance of the proposed MOCO method. Both data sets are collected by UAV SAR systems. Owing to the small size and light weight of the platform, trajectory deviations are easily introduced by the presence of atmosphere turbulence.

The first real data set is recorded in highly squinted mode, and its trajectory parameters are measured by an IMU. Despite the fact that the IMU data is slightly inaccurate in terms of focusing, it can be used for performance comparison. Detailed analyses in this part include LPEFs (estimated with and without smoothness regularization, respectively), residual trajectory deviations (obtained through LS, WLS, TLS, and the proposed WTLS, respectively), and the full-swath imaging results (autofocused by the raw trajectory, the conventional PGA [6], the WLS- and WTLS-corrected trajectory, respectively).

The second experiment is performed for large-swath SAR reconstruction, in which the space variance of the motion error is more pronounced. Results comparison will be provided via the images generated by the weighted PGA (WPGA) [8] with blocking in both range and cross-range directions [4], [5] and the proposed MOCO, respectively.

All experiments are carried out on a personal computer with Intel Core i7 CPU at 3.6 GHz and 8-GB RAM.

Date-related information (i.e., system and image parameters) is resumed in Table I.

### A. High Squint Example

After some necessary preprocessing for the IMU data, including time alignment, filtering, resampling, and coordinate transform [39], the measured trajectory is given in Fig. 5. It is obvious that large deviations exist in three motion directions. Partial trajectory is extracted from the IMU data, and the unused trajectory deviations in Fig. 6 are considered as “unknown” motion errors, which will be compared with the result of the proposed MOCO in the following. Although the IMU data is not sufficiently accurate for focusing, it can be

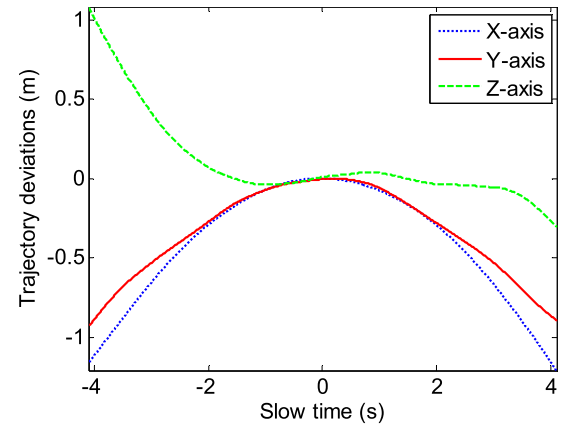


Fig. 5. Complete trajectory deviations from IMU.

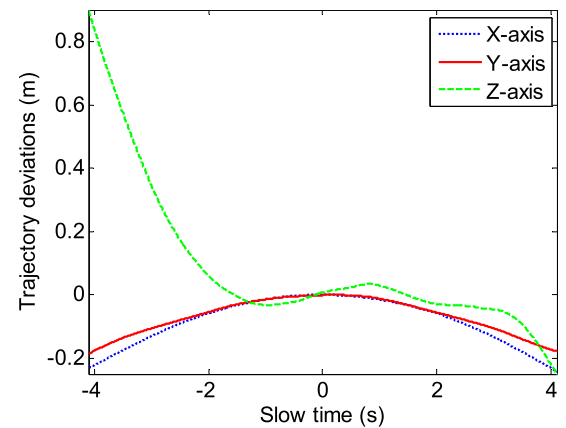


Fig. 6. Unused trajectory deviations.

used for performance comparison. If the estimated trajectory inherits the trend of the IMU-measured one, but provides better focusing performance, the effectiveness of the proposed MOCO approach can be validated.

Using the partial track parameters, the subsequent FFBP image is depicted in Fig. 7(a), which is seriously defocused. On the contrary, the image generated by the complete trajectory in Fig. 5 is shown in Fig. 7(b), which is focused visually, but contains slight blur from our magnified observation. This implies that minor motion error still exists in the IMU measurements. For comparison, the blurred image in Fig. 7(a) is then autofocused by the conventional PGA

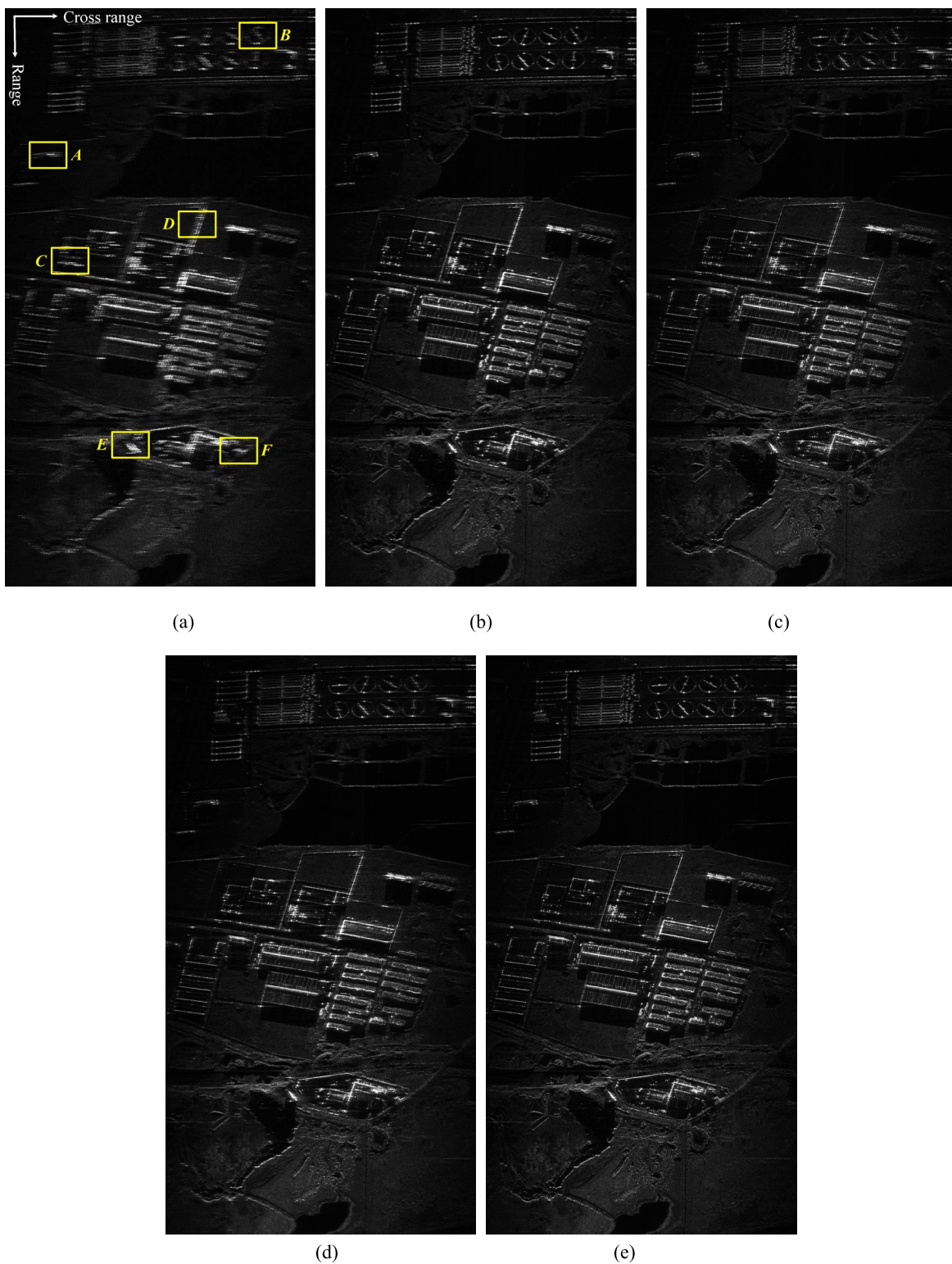


Fig. 7. Image results. (a) Image with partial IMU data. (b) Image with complete IMU data. (c) Image autofocused by PGA. (d) Image autofocused by the WLS-based MOCO. (e) Image autofocused by the WTLS-based MOCO.

method [6]. PGA is a postprocessing step, and only able to correct the space-invariant phase errors [6]. Therefore, MOCO via PGA remains incomplete, exhibiting visible blurs in Fig. 7(c). Fig. 7(d) shows the image autofocused by the WLS-based MOCO, and Fig. 7(e) is the one obtained by the

WTLS-based MOCO. It can be seen that Fig. 7(e) is more concentrated than Fig. 7(a)–(d). To make a clearer comparison, the relative sharpness, entropy and contrast values for Fig. 7 are computed and listed in Table II, which shows that Fig. 7(e) achieves the best focusing performance.

TABLE II  
PERFORMANCE VALUES COMPARISON FOR FIG. 7

| Image              | (a)   | (b)   | (c)   | (d)   | (e)   |
|--------------------|-------|-------|-------|-------|-------|
| Relative sharpness | 1.00  | 5.59  | 8.53  | 4.16  | 19.12 |
| Entropy            | 12.28 | 11.71 | 11.60 | 11.78 | 11.32 |
| Contrast           | 1.74  | 1.91  | 1.94  | 1.88  | 2.17  |

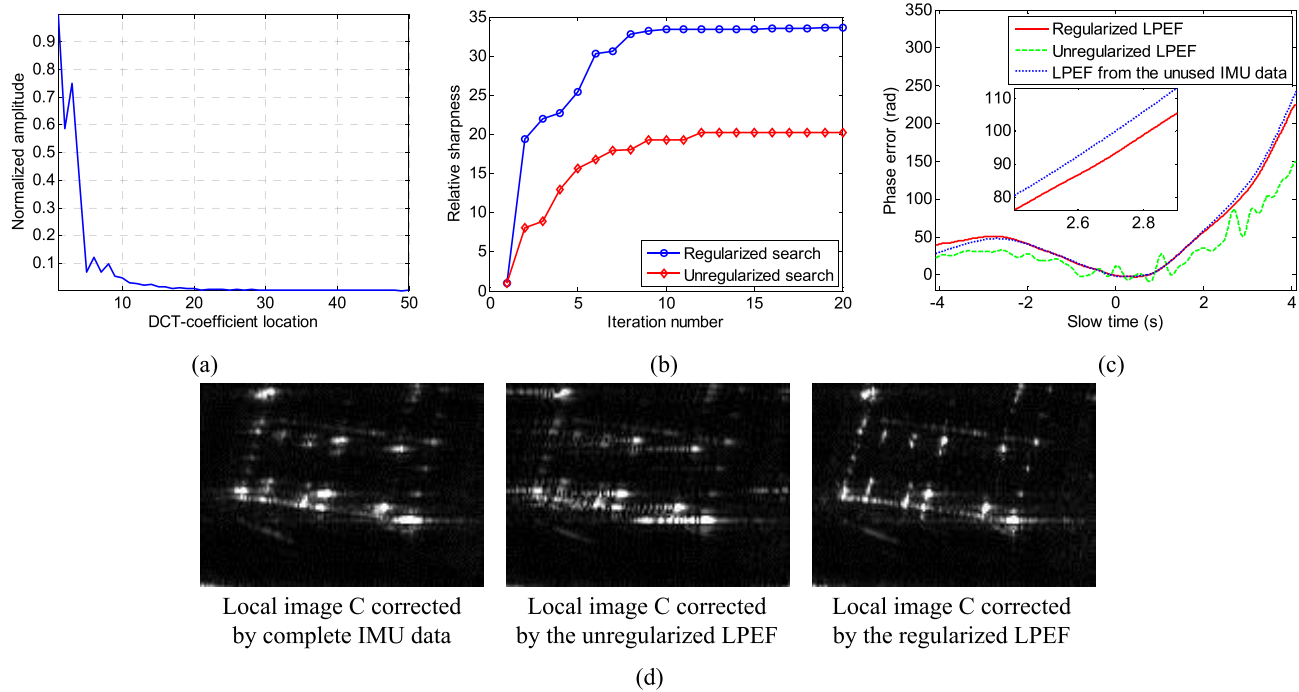


Fig. 8. LPEF estimation for local image C. (a) Regularization vector. (b) Relative sharpness value versus the iteration number in BFGS search. (c) LPEFs from the regularized solution, the unregularized solution, and the unused trajectory deviations in Fig. 6, respectively. (d) Local image C corrected by complete IMU data, the unregularized LPEF, and the regularized LPEF, respectively.

To clarify the details of the proposed MOCO method, six local images highlighted by the rectangles A-F in Fig. 7(a) are selected. The size of individual local image is set as  $100 \times 140$  (range  $\times$  angle) pixels to guarantee the space invariance of the LPEF. Obviously, the selected local images contain a large amount of strong points, cover the most dispersed power, and distribute over a wide extent. For LPEFs estimation, the number of DCT-coefficients is limited to be 50, which is accurate enough for our experience. The smoothness regularization is utilized to enhance the low-frequency drift of the DCT-coefficients, and the regularized vector is determined by the IMU data.

Taking local image C for example, the imperfect LPEF is first calculated by (2) according to the used trajectory deviations. The regularized vector can be obtained by retaining the first 50 elements of the DCT distribution of the imperfect LPEF, as illustrated in Fig. 8(a). Note again, the main purpose of adding smoothness regularization to the gradient formula is to emphasize the low-frequency drift of the DCT distribution, and the inaccuracy of the regularization vector can be made up by the BFGS iterations. The sharpness metric with respect to the iteration number is plotted in Fig. 8(b). As shown in this figure, adding smoothness regularization achieves higher cost

value in comparison with the unregularized solution, which indicates that the unregularized search is probably trapped in a local maximum. This can be confirmed by the estimated LPEFs in Fig. 8(c), where the red solid line is the regularized solution, the green dashed line comes from the unregularized search, and the blue dotted line is calculated according to the unused IMU data in Fig. 6. The advantage of considering smoothness regularization is manifested in Fig. 8(c), where the regularized LPEF inherits the trend of the IMU measurements, and the unregularized solution contains large high-frequency disturbance on the contrary. The minor difference between the regularized LPEF and the IMU-measured one indicates the accuracy limitation of the IMU sensor. By correcting three LPEFs in Fig. 8(c), the corresponding local images of area C are presented in Fig. 8(d), which shows evidence that the regularized LPEF gives the best focusing performance.

All six local images are autofocused by the sharpness-maximizing optimization, and the derived LPEFs are plotted in Fig. 9(a). The space variance of the LPEFs is noticeable from the magnified illumination in Fig. 9(b). As shown in Fig. 9, the phase error is small at the center of aperture and increase with the slow time, so does the space variance of LPEFs. This reflects that the trajectory deviations become

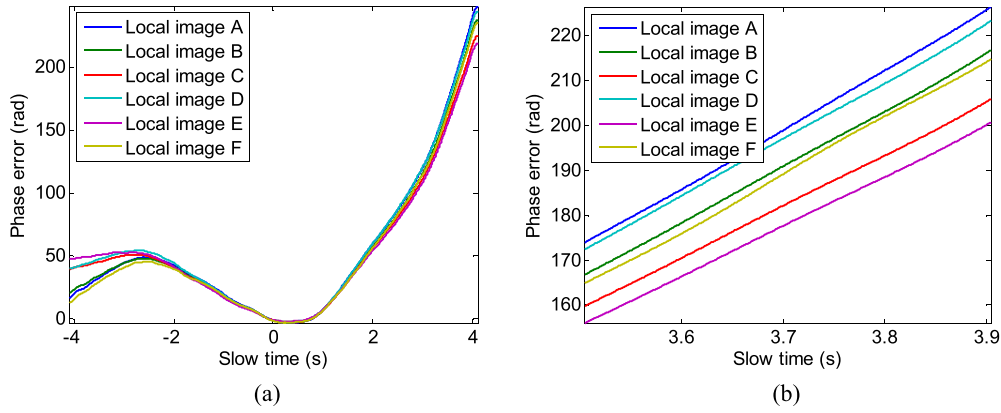


Fig. 9. Estimated LPEFs. (a) LPEFs for local images A–F. (b) Zoomed-in view of Fig. 9(a).

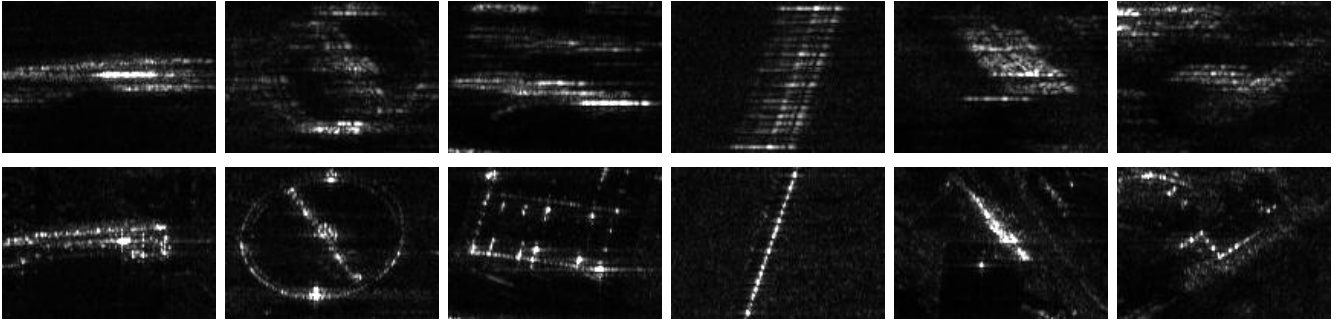


Fig. 10. Local images comparison. Before (top row) and after (bottom row) sharpness maximum autofocusing.

TABLE III  
PERFORMANCE VALUES BEFORE AND AFTER LOCAL IMAGES AUTOFOCUSING FOR FIG. 10

| Local image        |                     | A     | B     | C     | D     | E     | F    |
|--------------------|---------------------|-------|-------|-------|-------|-------|------|
| Relative sharpness | Before autofocusing | 1     | 1     | 1     | 1     | 1     | 1    |
|                    | After autofocusing  | 14.20 | 45.96 | 33.59 | 27.33 | 16.65 | 4.69 |
| Entropy            | Before autofocusing | 7.53  | 5.92  | 6.81  | 7.69  | 7.35  | 7.44 |
|                    | After autofocusing  | 6.31  | 3.93  | 4.42  | 5.94  | 5.15  | 6.73 |
| Contrast           | Before autofocusing | 1.32  | 2.58  | 1.97  | 1.23  | 1.55  | 1.39 |
|                    | After autofocusing  | 1.90  | 3.57  | 3.31  | 1.82  | 2.29  | 1.57 |

more and more pronounced with the increase of slow time. The defocused and LPEF-corrected local images are illustrated in Fig. 10, and their relative sharpness, entropy and contrast values are listed in Table III. A clear focusing improvement can be observed from the results in Fig. 10 and Table III, which ensures the success of the sharpness-maximizing optimization.

Next, the residual trajectory deviations are estimated according to the LPEFs. The solutions of LS, WLS, TLS and WTLS are shown in Fig. 11, respectively. As shown in this figure, the LS, WLS, and TLS results contain either large jumps or errors far away from the IMU data. Distinctively, the proposed WTLS estimator is highly consistent with the trend of the platform movement, but exhibits some high-frequency motion information. This reflects the accuracy limitation of the IMU sensor, and slight blur is induced in Fig. 7(b) by these unrecorded high-frequency motion errors. By updating the trajectory parameters, the precision of the WTLS estimator

can be demonstrated by the FFBP image, as shown in Fig. 7(e). In addition, the total runtime of LS, WLS, TLS, and WTLS are 0.47, 0.48, 0.19, and 4.21 s, respectively. In spite of the highest computational burden, the WTLS method provides the trajectory parameters of best precision.

To clarify the focusing enhancement, two magnified subimages at near and far range are depicted in Fig. 12, respectively. As shown in Fig. 12, blurs still exist in the sub-images processed by PGA and the WLS-based MOCO, while the error disturbance is almost completely eliminated in the proposed MOCO results. This can be confirmed by the measured sharpness, entropy, and contrast values, as listed in Table IV. To make a quality assessment, the angular profiles of two point-like targets, P1 at near range and P2 at far range, are plotted in Fig. 13, and the related peak sidelobe ratio (PSLR), integrated sidelobe ratio (ISLR), and impulse response width (IRW) [5] are listed in Table V. It can be seen that all quality parameters derived by the proposed

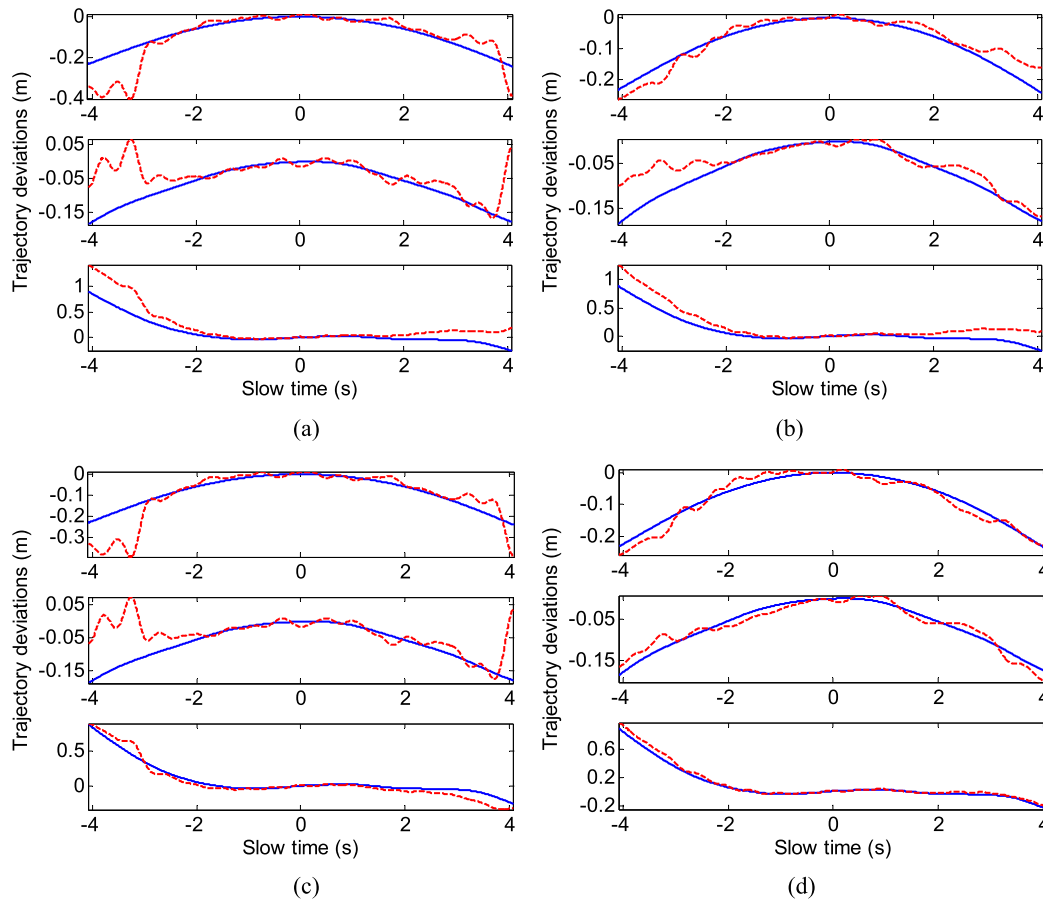


Fig. 11. Estimated trajectory deviations by (a) LS, (b) WLS, (c) TLS, and (d) WTLS, respectively. Blue solid line: deviations from the unused IMU data. Red dashed line: estimation result. In each figure, the top, middle, and bottom plots represent the deviations in  $x$ -,  $y$ -, and  $z$ -directions, respectively.

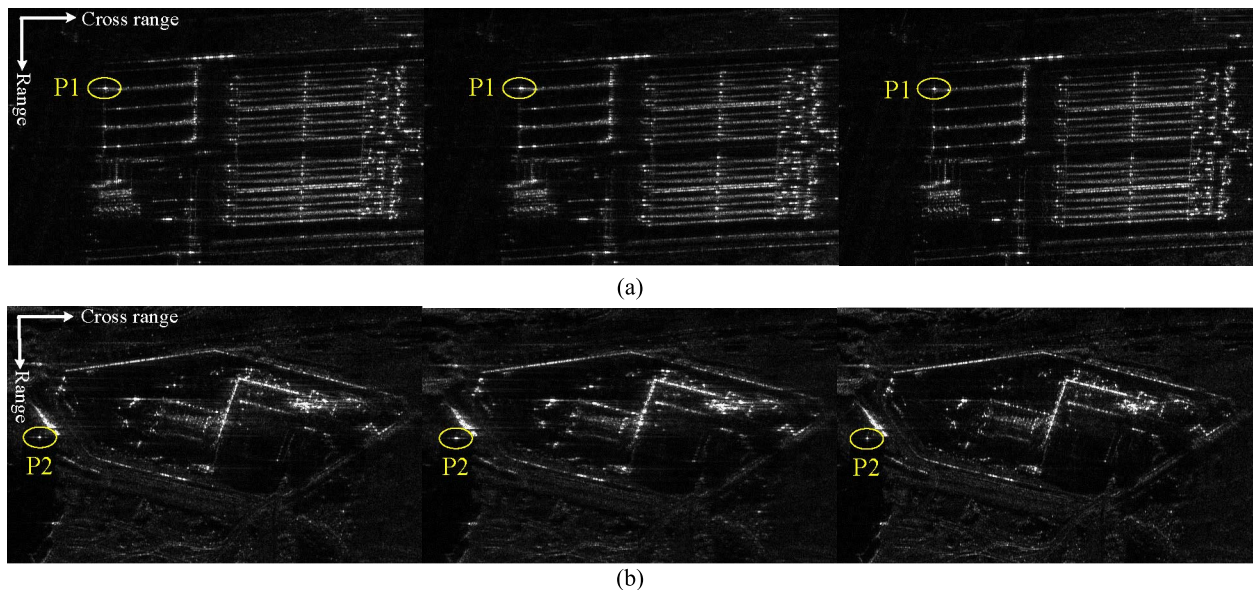


Fig. 12. Subimages comparison. (a) Subimages at near range obtained by PGA (left), WLS-based MOCO (middle), and WTLS-based MOCO (right). (b) Subimages at far range generated by PGA (left), WLS-based MOCO (middle), and WTLS-based MOCO (right).

MOCO outperform the ones using PGA and the WLS solution. In terms of time efficiency, the total runtime of PGA and the proposed MOCO is 2.7 and 23.4 min, respectively. Since the

proposed MOCO is computationally expensive, it is recommended to accelerate the processing routine by parallelizing implementation in practice.

TABLE IV  
PERFORMANCE VALUES COMPARISON FOR FIG. 12

| Sub-image          | (a)  |                |                 | (b)  |                |                 |
|--------------------|------|----------------|-----------------|------|----------------|-----------------|
|                    | PGA  | WLS-based MOCO | WTLS-based MOCO | PGA  | WLS-based MOCO | WTLS-based MOCO |
| Relative sharpness | 1.00 | 0.68           | 1.09            | 1.00 | 0.49           | 1.13            |
| Entropy            | 9.83 | 9.97           | 9.77            | 9.16 | 9.43           | 9.05            |
| Contrast           | 1.71 | 1.66           | 1.79            | 1.76 | 1.69           | 1.88            |

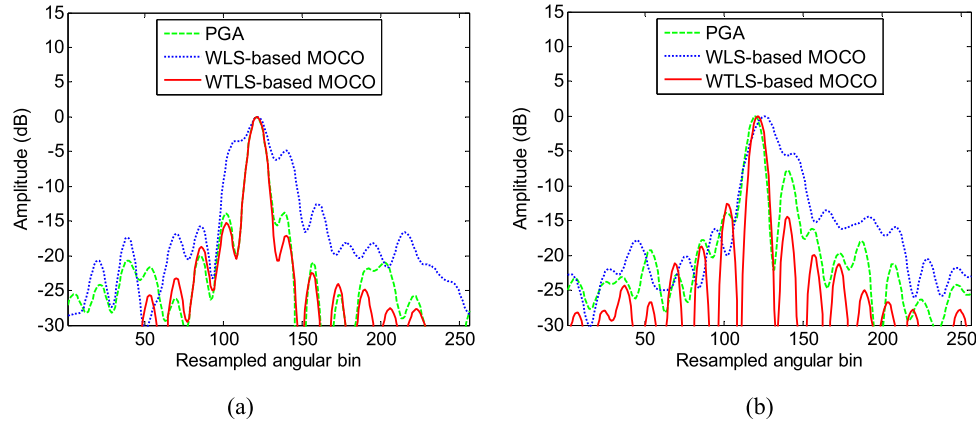


Fig. 13. Angular profiles. (a) P1 at near range. (b) P2 at far range.

TABLE V  
QUALITY PARAMETERS COMPARISON FOR FIG. 13

| Target             | P1     |                |                 | P2    |                |                 |
|--------------------|--------|----------------|-----------------|-------|----------------|-----------------|
|                    | PGA    | WLS-based MOCO | WTLS-based MOCO | PGA   | WLS-based MOCO | WTLS-based MOCO |
| PSLR(dB)           | -13.82 | -3.45          | -15.31          | -7.79 | -5.40          | -12.64          |
| ISLR(dB)           | -8.98  | -2.20          | -11.40          | -5.11 | -5.92          | -10.73          |
| IRW(angular grids) | 1.05   | 1.75           | 1.04            | 1.14  | 1.88           | 1.02            |

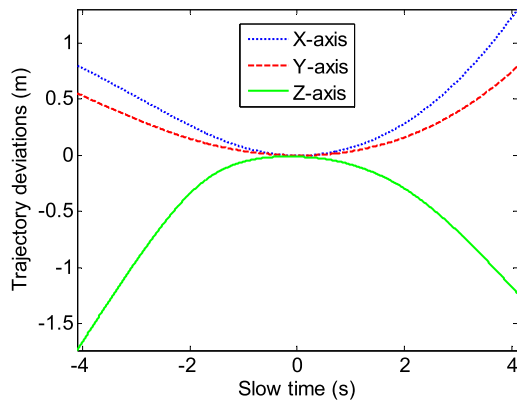


Fig. 14. Raw trajectory from IMU.

### B. Large Swath Example

In this part, the proposed MOCO performance will be investigated by the second data set in large-swath mode. Data-related parameters can be found in Table I.

The raw trajectory deviations from IMU are given in Fig. 14, and the subsequent FFBP image is shown in Fig. 15(a). It can be observed that Fig. 15(a) is roughly focused, and some detailed ground information is even invisible. This reflects

the low accuracy of the recorded trajectory, which makes it necessary to estimate motion errors from the SAR echoes.

To make a focusing comparison, the WPGA [8], [27] and the proposed MOCO are applied to generate the refocused image, respectively. WPGA is an effective candidate for airborne SAR autofocus by weighting the phase error functions of multiple isolated prominent points. To eliminate the space-variant effect of motion error, the raw data is firstly divided into small blocks in both range and cross-range directions, then the WPGA can be path-wise applied. Fig. 15(b) shows the image autofocused by the WPGA-based MOCO. Fig. 15(b) is visually focused, but exhibiting obvious blurs in some boundary area. The image obtained by the proposed MOCO is given in Fig. 15(c), which is more concentrated than the WPGA-based result.

To implement the proposed autofocus method, four local images marked by A-D in Fig. 15(a) are selected and autofocused under the criterion of maximum sharpness. Each local image contains  $200 \times 300$  (range  $\times$  angle) pixels, and the related LPEF is approximated by 30-order DCT coefficients. The estimation results by the BFGS search are depicted in Fig. 16, where the space variance of LPEFs is noticeable in large-swath mode. The local images before and after sharpness-maximizing autofocusing are presented in Fig. 17, and their

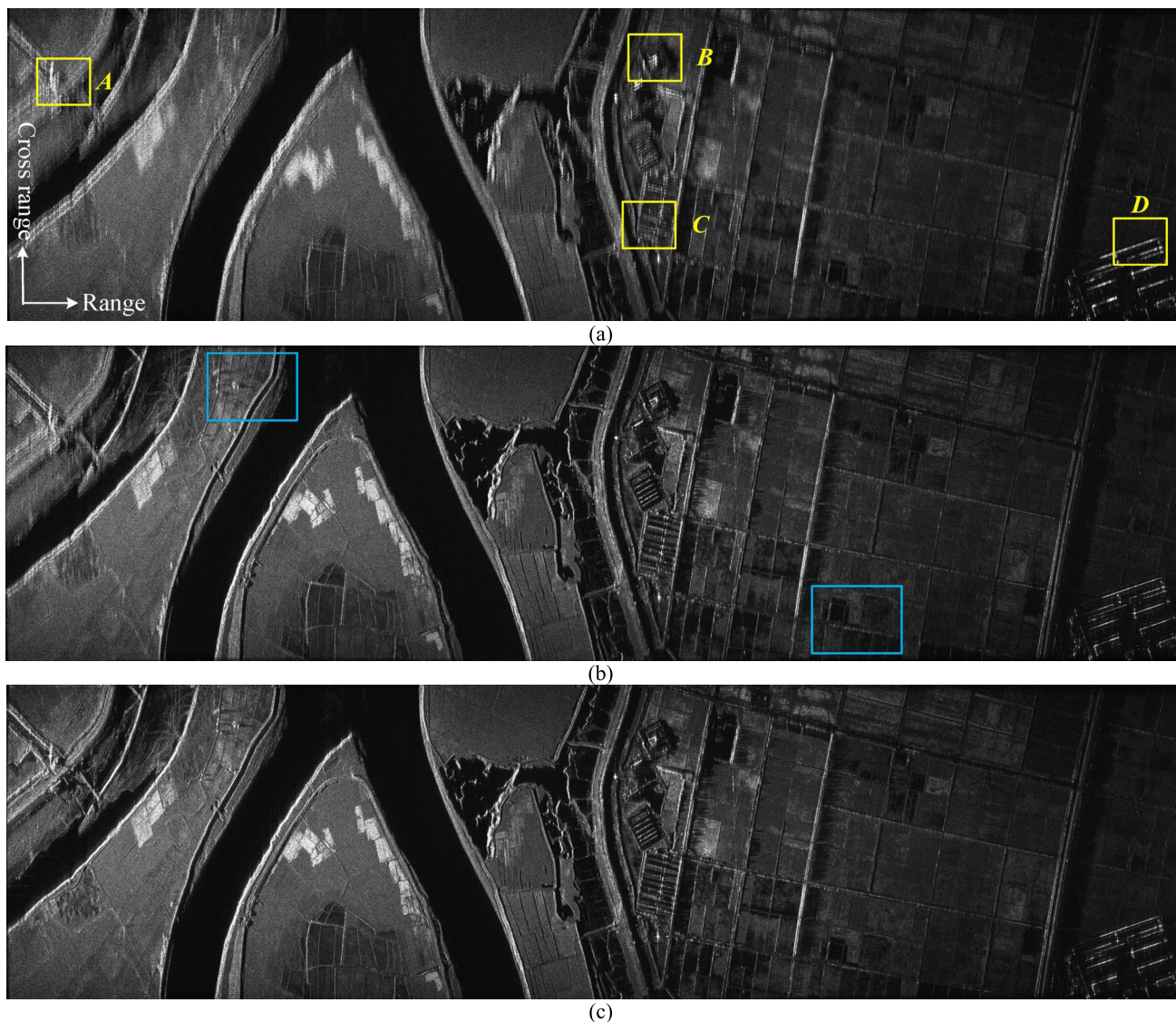


Fig. 15. Images generated by (a) IMU-measured trajectory, (b) WPGA-based MOCO, and (c) proposed MOCO.

TABLE VI  
PERFORMANCE VALUES BEFORE AND AFTER LOCAL  
IMAGES AUTOFOCUSING FOR FIG. 17

| Local image        |                     | A    | B    | C    | D    |
|--------------------|---------------------|------|------|------|------|
| Relative sharpness | Before autofocusing | 1    | 1    | 1    | 1    |
|                    | After autofocusing  | 6.87 | 6.14 | 6.51 | 3.38 |
| Entropy            | Before autofocusing | 8.17 | 9.39 | 9.02 | 8.43 |
|                    | After autofocusing  | 7.35 | 8.88 | 8.20 | 7.82 |
| Contrast           | Before autofocusing | 1.28 | 0.90 | 1.11 | 1.46 |
|                    | After autofocusing  | 1.39 | 1.07 | 1.32 | 1.61 |

sharpness, entropy and contrast values are listed in Table VI. It is visually evident that the LPEF-corrected local images have achieved better focusing performance. By combining the LPEFs into the WTLS model, the derived 3-D trajectory deviations are shown in Fig. 18. We can see that the residual maximum deviation is high up to the order of decimeter, which is far larger than the wavelength of the transmitted signal in Ku-band. By adding the raw trajectory in Fig. 14 and the

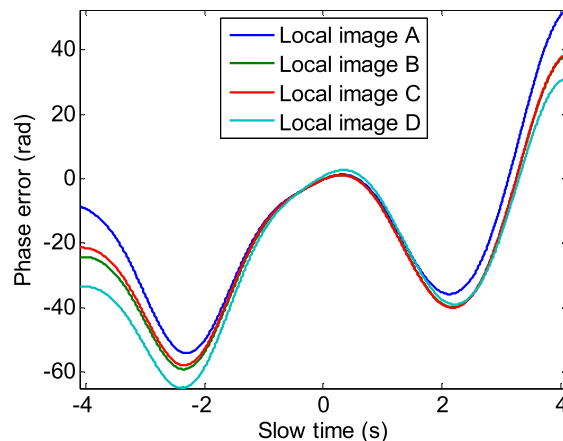


Fig. 16. Estimated LPEFs for four local images.

estimated one in Fig. 18, final trajectory is now available for precise SAR reconstruction, generating the focused image in Fig. 15(c).

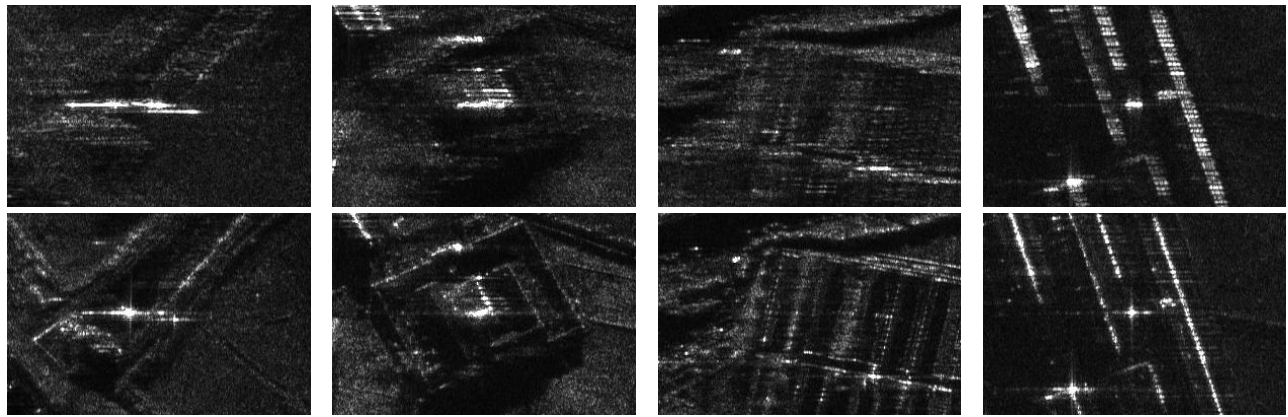


Fig. 17. Local images before (top row) and after (bottom row) sharpness maximum autofocusing.

TABLE VII  
PERFORMANCE VALUES COMPARISON FOR FIG. 19

| Sub-image          | (a)   |               | (b)   |               |
|--------------------|-------|---------------|-------|---------------|
|                    | WPGA  | Proposed MOCO | WPGA  | Proposed MOCO |
| Relative sharpness | 1.00  | 1.28          | 1.00  | 1.12          |
| Entropy            | 11.74 | 11.69         | 11.87 | 11.83         |
| Contrast           | 0.63  | 0.67          | 0.54  | 0.59          |

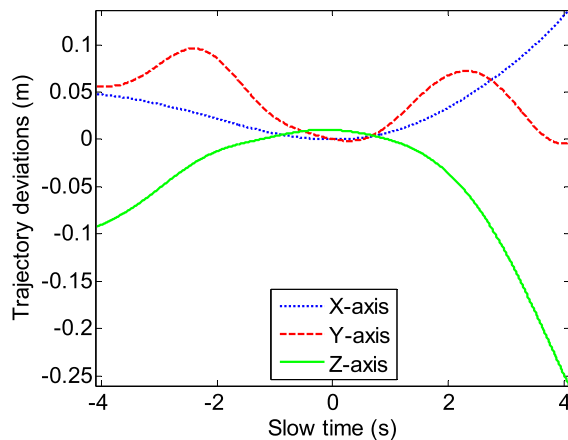


Fig. 18. Estimated trajectory deviations.

For clarity, two subimages highlighted by the rectangular domains in Fig. 15(b) are amplified and presented in Fig. 19. In WPGA-based subimages, some detailed ground information is missing. Specifically, the trees and the texture of some farmland are blurred, which may be caused by the lack of prominent points for the phase gradient estimation in WPGA. On the contrary, all these scenes can be clearly distinguished in the proposed MOCO results. The relative sharpness, entropy, and contrast values for Fig. 19 are measured and listed in Table VII, which further demonstrates the focusing enhancement of the proposal over the WPGA-based result. In addition, the runtime of the WPGA-based MOCO is about 6.2 min, while it takes 24.9 min for the proposed MOCO. We observe that the sharpness-maximum optimization is more time-consuming than the WTLS solver; works on decreasing

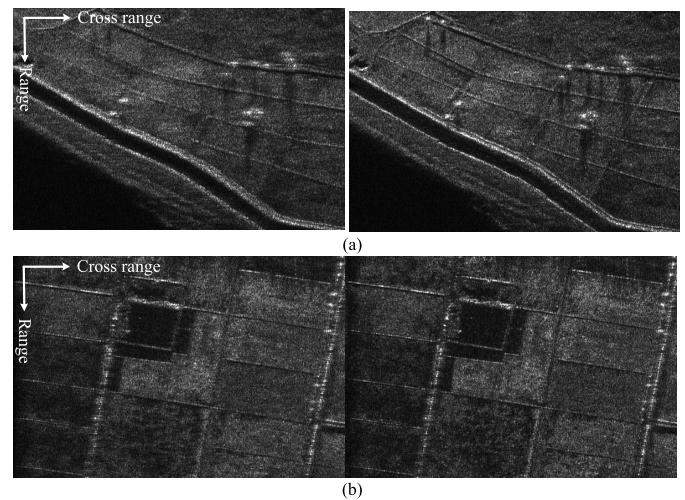


Fig. 19. Amplified subimages. (a) Subimages obtained by WPGA-based MOCO (left) and the proposed MOCO (right). (b) Subimages generated by WPGA-based MOCO (left) and the proposed MOCO (right).

the computational burden for the proposed MOCO should be further considered in the future.

## VII. CONCLUSION

A new autofocus algorithm has been proposed for airborne SAR imagery in this study, which improves the capacity to estimate the residual 3-D trajectory deviations. The motion error model compatible with the BP formation is presented. During the autofocus processing, multiple local images at different illumination directions are first selected and refocused under the criterion of maximum sharpness, generating related LPEFs. By combining the LPEFs into a linear system of

equations, the residual trajectory deviations can be retrieved through a proposed WTLS method. And finally, the image can be focused using the updated trajectory. A number of real data experiments validate the effectiveness of the proposed approach.

## APPENDIX

At the beginning of the BFGS search, the initial value of the Hessian matrix is set as  $\mathbf{B}^{[0]} = \mathbf{I}$ , where  $\mathbf{I}$  is an identity matrix. To guarantee the symmetry positive-definiteness of the Hessian matrix in each iteration of the BFGS search,  $\mathbf{B}^{[u]}$  is estimated based on the following rule [32]:

$$\mathbf{B}^{[u+1]} = \begin{cases} \mathbf{B}^{[u]}, & \text{if } ((\mathbf{y}^{[u]})^T \mathbf{s}^{[u]} \leq 0) \\ \mathbf{B}^{[u]} - \frac{\mathbf{B}^{[u]} \mathbf{s}^{[u]} (\mathbf{s}^{[u]})^T \mathbf{B}^{[u]}}{(\mathbf{s}^{[u]})^T \mathbf{B}^{[u]} \mathbf{s}^{[u]}} + \frac{\mathbf{y}^{[u]} (\mathbf{y}^{[u]})^T}{(\mathbf{y}^{[u]})^T \mathbf{s}^{[u]}}, & \text{if } ((\mathbf{y}^{[u]})^T \mathbf{s}^{[u]} > 0). \end{cases} \quad (34)$$

where the superscript  $u$  is the iterative number,  $\mathbf{s}^{[u+1]} = \hat{\boldsymbol{\phi}}_d^{[u+1]} - \hat{\boldsymbol{\phi}}_d^{[u]}$ , and  $\mathbf{y}^{[u+1]} = \nabla f_{\text{reg}}(\hat{\boldsymbol{\phi}}_d^{[u+1]}) - \nabla f_{\text{reg}}(\hat{\boldsymbol{\phi}}_d^{[u]})$ .

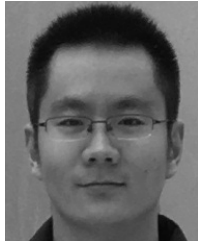
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